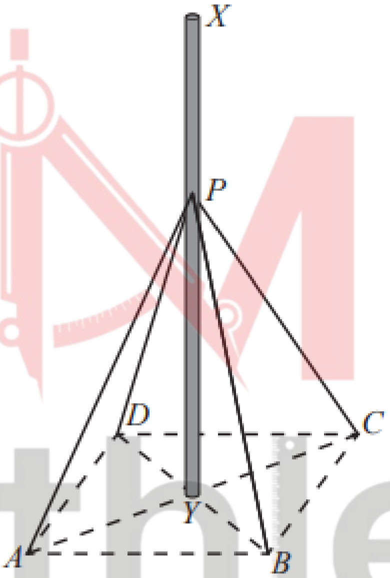
# Trigonometry

## Worksheet 4a

### 3-D Trigonometry

1. **W17/qp22/q11**




A vertical mast, XY, is positioned on horizontal ground. The mast is supported by four cables attached to the mast at P and to the ground at points A, B, C and D. Y is the centre of the square ABCD. PY = 7.50m.

a. Given that AB = 3.65m, show that AY = 2.58m correct to 3 significant figures. [3]

b. Calculate the length of one of the cables used to support the mast. [2]

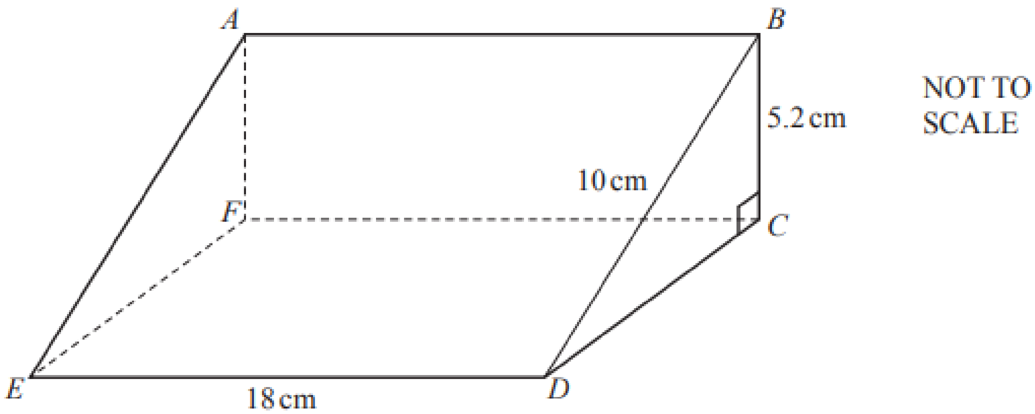
c. Calculate $A\hat{P}B$. [3]

The angle of elevation of X from A is 77.0°.

d. Calculate the height, XY of the mast. [2]

e. Calculate the angle of elevation of X from the midpoint of AB. [2]

## 2. W19/qp42/q4(b)(i)(b) and (ii) (0580)

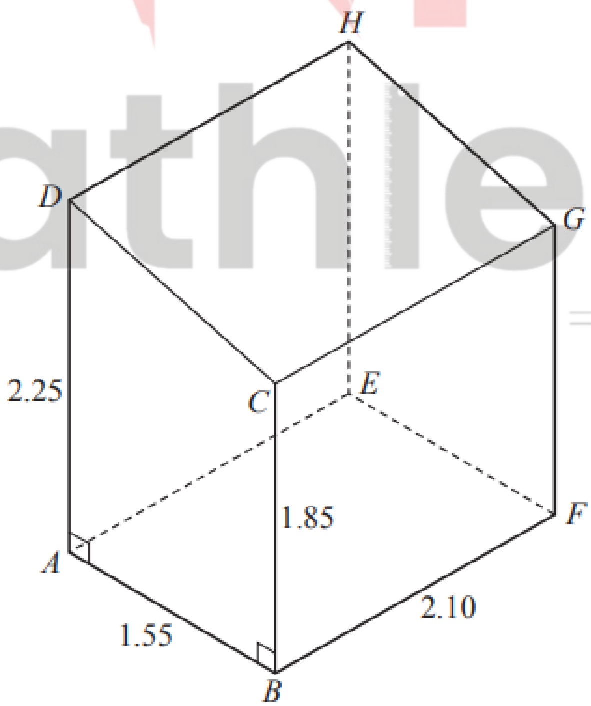



The diagram shows a prism ABCDEF. The cross-section is a right-angled triangle BCD. BD = 10cm, BC = 5.2cm and ED = 18cm.

i. Calculate angle BEC. [4]

ii. The point G lies on the line ED and GD = 7cm. Work out angle Calculate $B\hat{G}E$. [3]

## 3. W20/qp21/q5(c)

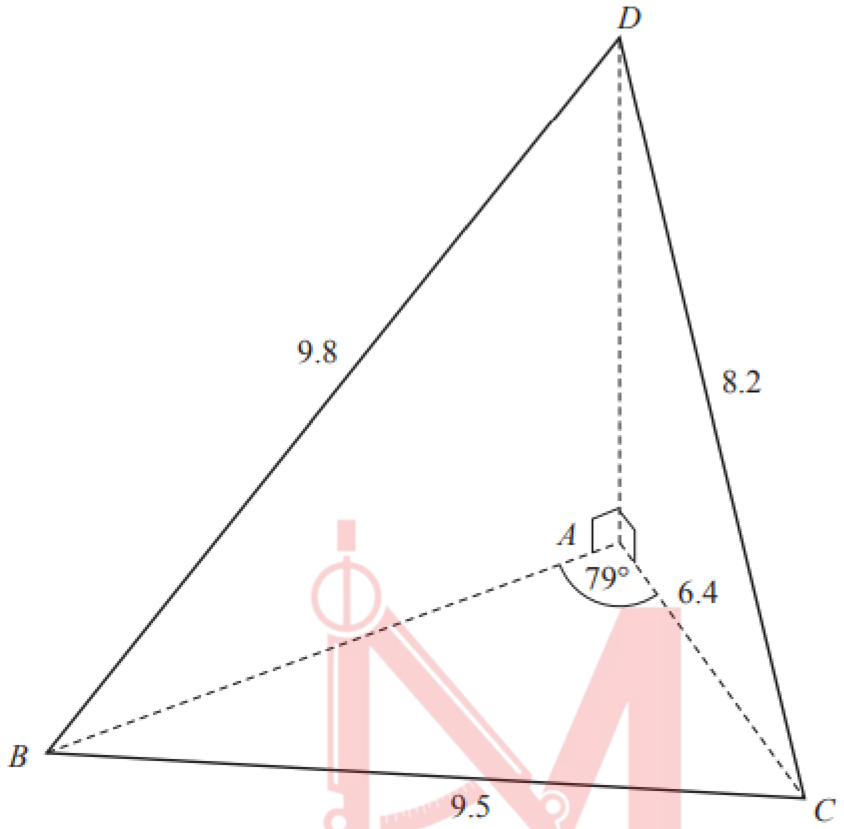





The diagram shows a garden shed positioned on horizontal ground. It is in the shape of a prism with trapezium ABCD as its cross-section

Calculate the angle of elevation of D from F. [4]

### 4. W20/qp22/q9(b)

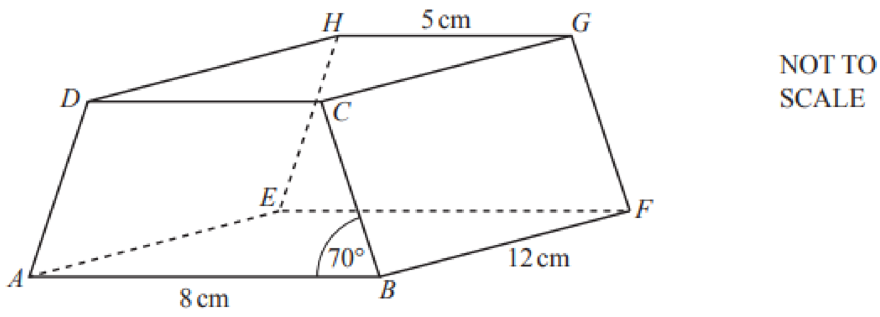



In triangle ABC, AC = 6.4cm, BC = 9.5cm and BAC = 79°. The same triangle ABC forms the horizontal base of a pyramid ABCD. BD = 9.8cm and CD = 8.2cm. $B\hat{A}D = C\hat{A}D = 90°$.

i. Calculate $B\widehat{D}C$. [3]

ii. Calculate the angle of elevation of D from C. [2]

**5. W20/qp42/q9(b) (0580)**




The diagram shows a prism with a rectangular base, ABFE. The cross-section, ABCD, is a trapezium with AD = BC. AB = 8cm, GH = 5 cm, BF = 12cm and angle ABC = 70°.

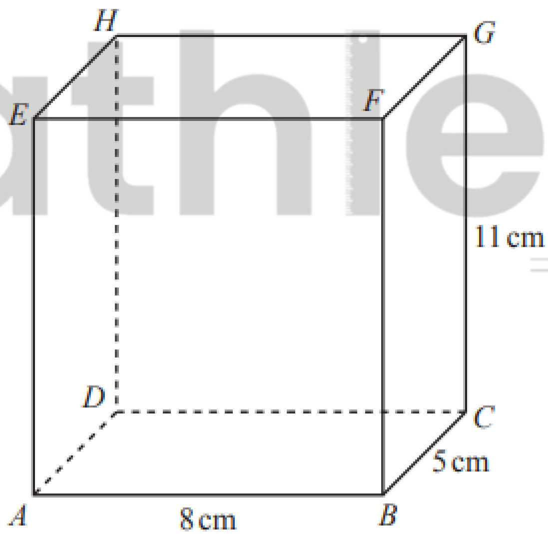
The perpendicular from G onto EF meets EF at X.

i. Show that EX = 6.5cm. [1]

ii. Calculate AX. [2]

iii. Calculate the angle between the diagonal AG and the base ABFE. [2]

**6. W20/qp43/q6(b)(c) (0580)**



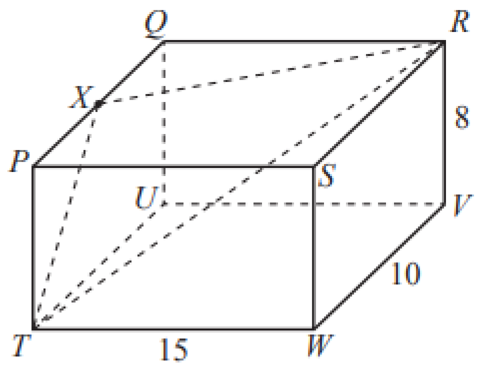

a. Ivana has a pencil of length 13cm. Does this pencil fit completely inside the cuboid? Show how you decide. [4]

b.

i. Calculate angle $C\hat{A}B$. [2]

ii. Calculate angle $G\hat{A}C$. [2]

### 7. S21/qp21/q4(b)

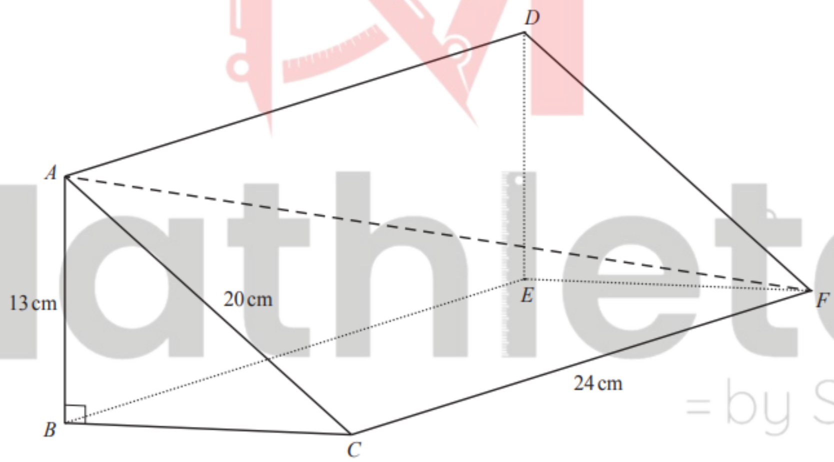



The diagram shows a cuboid. TW = 15cm, WV = 10cm and RV = 8cm.

i. Show that TR = 19.7cm, correct to 1 decimal place. [3]

ii. X is the midpoint of PQ. Calculate . $T\hat{R}X$. [5]

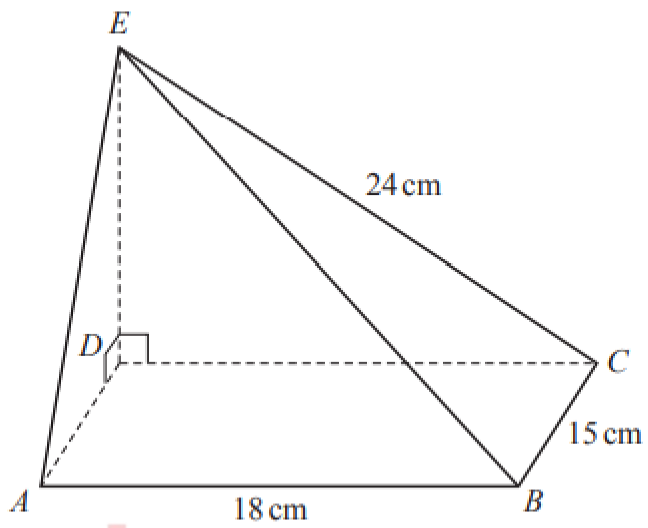
### 8. S21/qp41/q9(c) (0580)




The diagram shows a prism, ABCDEF. AB = 13cm, AC = 20cm, CF = 24cm and angle ABC = 90°.

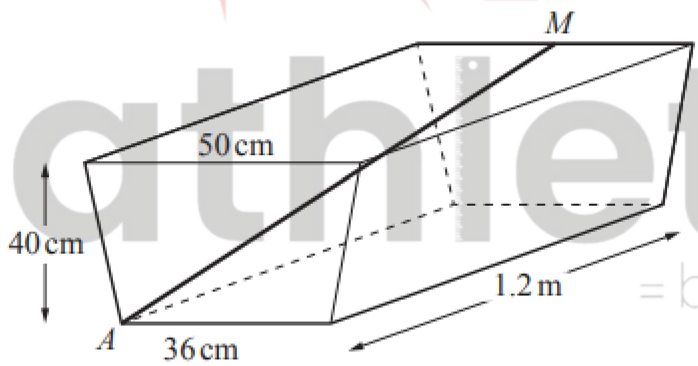
Calculate the angle that AF makes with the base BCFE. [4]

## 9. S21/qp42/q8(b)(ii) (0580)




The diagram shows a solid pyramid with a rectangular base ABCD. E is vertically above D. Angle EDC = angle EDA = 90°. AB = 18cm, BC = 15cm and EC = 24 cm. Calculate the angle between BE and the base of the pyramid. [4]

## 10. W22/qp43/q5(d) (0580)




A steel rod AM is placed inside the empty water trough as shown in the diagram. A is a vertex at the base of the isosceles trapezium and M is the midpoint of the top edge on the opposite face. Calculate the length of the steel rod, AM. [4]

## Answer Key

### 1. W17/qp22/q11

| | | | |
|---|---|---|---|
| 11(a) | Need to see 2.58 rounded from a correctly obtained 2 581 or better. | 3 | Method 1<br>**M2** for $AY = 3.65\cos 45$ or $(3.65 \div 2) \div \sin 45$<br>**or**<br>**M1** for e.g. $\frac{AY}{3.65} = \cos 45$ or $\sin 45 = \frac{3.65 \div 2}{AY}$<br><br>Method 2<br>**M1** for such as<br>$AY^2 + AY^2 = 3.65^2$ or $3.65^2 + 3.65^2 = AC^2$ soi<br><br>**M1** for $AY^2 = \frac{3.65^2}{2}$ oe<br><br>**A1** for $AY = 2.580[9\ldots]$ |
| 11(b) | 7.93 | 2 | **M1** for $7.5^2 + 2.58^2$ |
| 11(c) | 26.6° or $2\sin^{-1}\left(\frac{0.5 \times 3.65}{their 7.93}\right)$ | 3FT | **M2** for $2\sin^{-1}\left(\frac{0.5 \times 3.65}{their 7.93}\right)$<br>or $\cos[\ldots] = \frac{their 7.93^2 + their 7.93^2 - 3.65^2}{2 \times their 7.93^2}$<br>**Or**<br>**M1** for $\sin[\ldots] = \frac{0.5 \times 3.65}{their 7.93}$<br>or $3.65^2 = \text{their} 7.93^2 + \text{their} 7.93^2 - 2 \times \text{their } 7.93^2 \times \cos[\ldots]$ |
| 11(d)(i) | 11.18 or 11.2 | 2 | **M1** for $\tan 77 = \frac{XY}{2.58}$ oe |
| 11(d)(ii) | 80.7° | 2FT | **M1** for $\tan[\ldots] = \frac{their\ 11.2}{3.65 \div 2}$ |

## 2. W19/qp42/q4(b)(i)(b) and (ii)

| | | | |
|---|---|---|---|
| 4(b)(i)(b) | 14.6 or 14.62 to 14.63… | **4** | **M3** for $\sin BEC = \dfrac{5.2}{\sqrt{10^2+18^2}}$ oe<br>or **M2** for [$BE$=] $\sqrt{10^2+18^2}$ oe seen<br>or [$EC$ = ] $\sqrt{18^2+10^2-5.2^2}$ oe seen<br><br>or **M1** for [$BE^2$ =] $10^2 + 18^2$ oe seen<br>or [$EC^2$=] $18^2 + 10^2 - 5.2^2$ seen |
| 4(b)(ii) | 125 or 124.9 to 125.0… | **3** | **B2** for 55[.0…] seen<br>or **M2** for $180 - \tan^{-1}\left(\dfrac{10}{7}\right)$ oe<br><br>or $\cos EGB = \dfrac{11^2+(10^2+7^2)-(10^2+18)^2}{2\times11\times\sqrt{10^2+7^2}}$ oe<br><br>or **M1** for tan[ ] = $\left(\dfrac{10}{7}\right)$ oe<br><br>or for<br>$(10^2 + 18^2) = 11^2 + (10^2 + 7^2) - 2\times11\times$ $\sqrt{10^2+7^2}$ $\cos EGB$ oe |

## 3. W20/qp21/q5(c)

| | | | |
|---|---|---|---|
| 5(c) | 40.8 or 40.76… | **4** | **M2** for [$AF$ = ] $\sqrt{1.55^2+2.1^2}$ oe<br>or **M1** for [$AF^2$ = ] $1.55^2+2.1^2$ oe<br><br>AND<br>**M1** for tan[ ] = $\dfrac{2.25}{theirAF}$ oe soi |

## 4. W20/qp22/q9(b)

| | | | |
|---|---|---|---|
| 9(b)(i) | 63[.0] or 62.97... | **3** | **M2** for [cos =] $\dfrac{9.8^2+8.2^2-9.5^2}{2\times9.8\times8.2}$<br>or **M1** for<br>$9.5^2 = 9.8^2+8.2^2-2\times9.8\times8.2\times\cos[...]$ |
| 9(b)(ii) | 38.7 or 38.69... | **2** | **M1** for cos[...] = $\dfrac{6.4}{8.2}$ |

## 5. W20/qp42/q9(b) (0580)

| | | | |
|---|---|---|---|
| 9(b)(i) | 8 – ½ (8 – 5) or 5 + ½ (8 – 5) | **M1** | |
| 9(b)(ii) | 13.6 or 13.64 to 13.65 | **2** | **M1** for $12^2 + (6.5)^2$ oe |
| 9(b)(iii) | 16.8 or 16.9 or 16.79 to 16.91…<br>nfww | **2** | **M1** for identifying angle $GAX$ from a diagram or from working or better |

## 6. W20/qp43/q6(b)(c) (0580)

| | | | |
|---|---|---|---|
| 6(b) | $\sqrt{8^2+5^2+11^2}$ oe<br>or<br>$8^2 + 5^2 + 11^2$ and $13^2$<br><br><u>ALTERNATIVE</u><br>$\sqrt{8^2+11^2}$ or $8^2 + 11^2$ and $13^2$ | **M3** | **M2** for $8^2 + 5^2 + 11^2$ or $8^2 + 11^2$ oe<br>or **M1** for $8^2 + 5^2$ or $5^2 + 11^2$ oe |
| | Yes and 14.5 or 14.4 or 14.49…<br>or<br>Yes and 13.6[0…] | **A1** | Accept equivalent conclusion |
| 6(c)(i) | 32.0[…] | **2** | **M1** for tan[..] = $\frac{5}{8}$ oe |
| 6(c)(ii) | 49.4 or 49.38 to 49.39 | **2** | **M1** for sin[..] = $\frac{11}{\textit{their } AG}$ oe |

## 7. S21/qp21/q4(b)

| | | | |
|---|---|---|---|
| 4(b)(i) | $8^2 + 10^2 + 15^2$ oe | **M2** | **M1** for any correct 2D Pythag pair e.g. $8^2 + 10^2$ |
| | 19.72… | **A1** | |
| 4(b)(ii) | 28.[0] to 28.2 | **5** | **M1** for $8^2 + \left(\frac{10}{2}\right)^2$<br>**M1** for $15^2 + \left(\frac{10}{2}\right)^2$<br>**M2** for cos… = $\frac{19.7^2 + RX^2 - TX^2}{2 \times 19.7 \times RX}$<br>or **M1** for<br>$TX^2 = 19.7^2 + RX^2 - 2 \times 19.7 \times RX \cos\ldots$ |



## 8. S21/qp41/q9(c) (0580)

| | | | |
|---|---|---|---|
| 9(c) | 24.6 or 24.58 to 24.59 | **4** | **M3** for sin [...] = $\frac{13}{\sqrt{20^2+24^2}}$ oe<br>or **M2** for $\sqrt{20^2+24^2}$ or $\sqrt{24^2+20^2-13^2}$<br>or **M1** for $AF^2 = 20^2 + 24^2$ or $24^2 + 20^2 - 13^2$<br>or **M1** for correct angle identified |

## 9. S21/qp42/q8(b)(ii) (0580)

| | | | |
|---|---|---|---|
| 8(b)(ii) | 34.1 or 34.11 to 34.12 | **4** | **M3** for tan [ ] = $\frac{\sqrt{24^2-18^2}}{\sqrt{18^2+15^2}}$ oe<br>or **M2** for $\sqrt{18^2+15^2}$ isw<br>or $\sqrt{24^2+15^2}$ isw<br>or **M1** for $18^2 + 15^2$ isw or $24^2 + 15^2$ isw<br>or **M1** for indicating required angle is *EBD* |

## 10. W22/qp43/q5(d) (0580)

| | | | |
|---|---|---|---|
| 5(d) | 128 or 127.7 to 127.8 | 4 | **B3** for $40^2 + 120^2 + 18^2$ oe<br>OR<br>**B1** for horizontal length 18 soi<br>**M1** for any correct attempt at 2-dimensional Pythagoras' $18^2 + 120^2$, $120^2 + 40^2$, $18^2 + 40^2$ |